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High Orbit Rocket Engine Design



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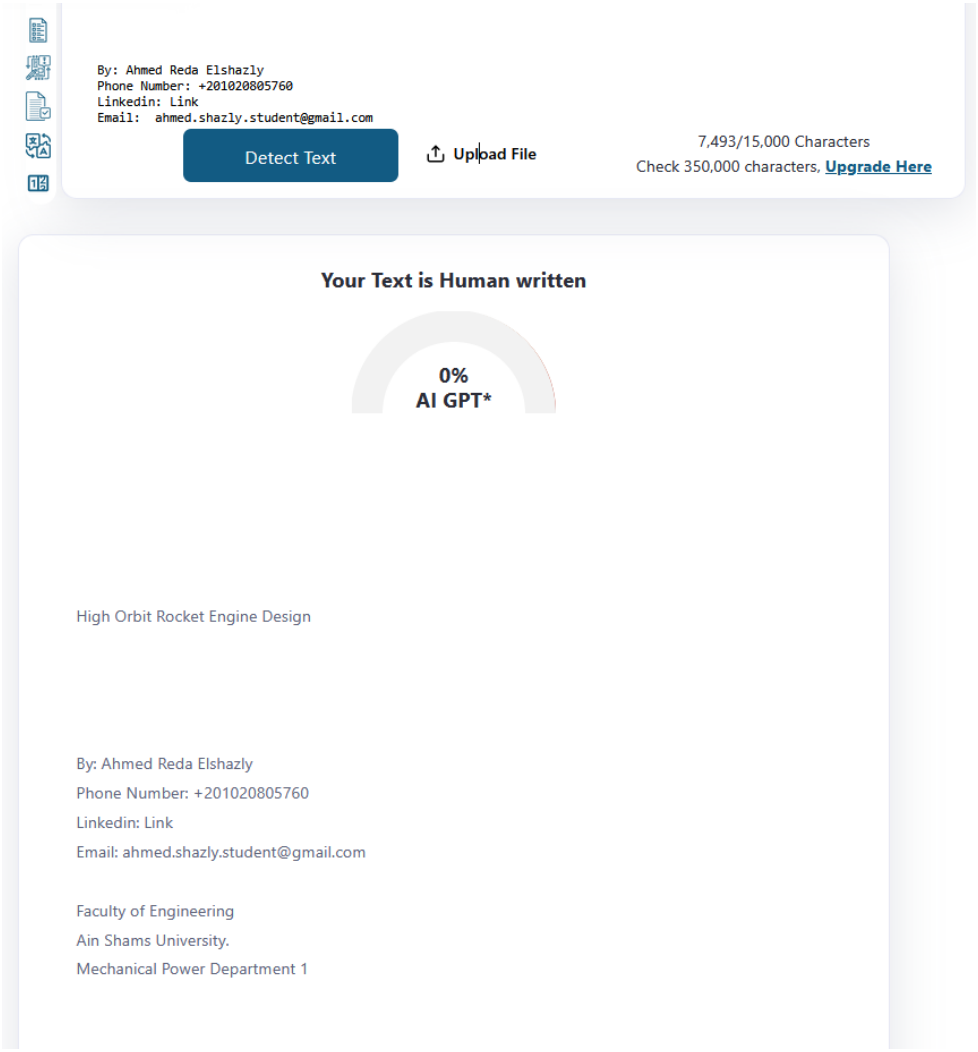
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Contents

Objectives 3
Design Specs of the Rocket Engine 4
Geometry Generation 5
Wall Contour using MOC in MATLAB 5
Modeling in Solidworks 6
Turbulence Model and Boundary Conditions 7
Boundary Conditions: 7
Notes 8
Convergence Criterion 8
Computational Domain 10
Results and post processing 11
Contours 11
Comparison with isentropic equation's calculations 13

Plagiarism

This document was not written by AI, all by my hand.



Objectives

•Generate the contour of engine walls using Minimum Length Method of Characteristics of Laplace equation for isentropic supersonic flow using MATLAB.

•Simulate the Rocket engine in ANSYS Fluent and validate the results of the team's EES code of the isentropic flow equations.

• Conducting a simulation of a high orbit rocket engine that's designed to carry 200-ton satellite to 1000KM in 5 mins.

Design Specs of the Rocket Engine

Design Parameters	Value
Mach number	4.687
K(mixture)	1.1973
R(mixture)	614.5
Mass of satellite	10000
Area Exit [m^2]	3.802060045
A*	0.049058839
Stagnation Temperature (T0) [K]	3580
Exit Temperature (T) [K]	1132.6313
Throat Temperature (T*) [K]	3258.54
Exit Velocity (Ve) [m/s]	4278.599814
Exit Pressure (Pa)	11500

Geometry Generation

Wall Contour using MOC in MATLAB

Used an opensource code of MOC that solves Laplace equation for supersonic flow to generate the diverging segment, edited the code to extract the coordinate points to generate Computational Domain. The code gets the maximum expansion angle via Prandtl Meyer Expansion Equation and generate the proper nozzle contour to taper off the expansion angle till the exit angle is, maximizing the velocity component in thrust's direction.

```
% Creat a 3-column matrix for solidworks (X, Y and Z=0)
% We use ' to transpose them into columns
solidworks_data = [x_wall',y_wall', zeros(size(x_wall'))];
%Save to a CSV file in your current folder
writematrix(solidworks_data, 'NozzleContour4.csv');

toc
```

Figure 1:Coordinate extractor command

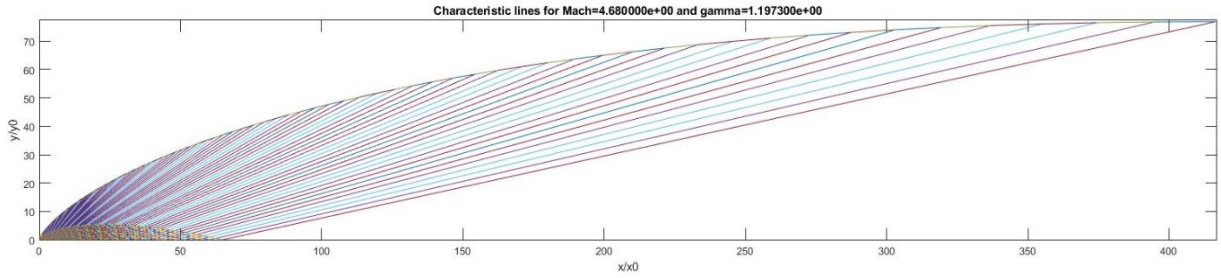


Figure 2:Wall Contour

Modeling in Solidworks

Since the problem is axis-symmetric, surface model of half longitudinal section of the Nozzle was generated in Solidworks. Divided the domain into Converging, Throat, Diverging, Mach Diamonds and Farfield regions to be able to control mesh generation.
Proper dimension for the ambient was selected as shown.

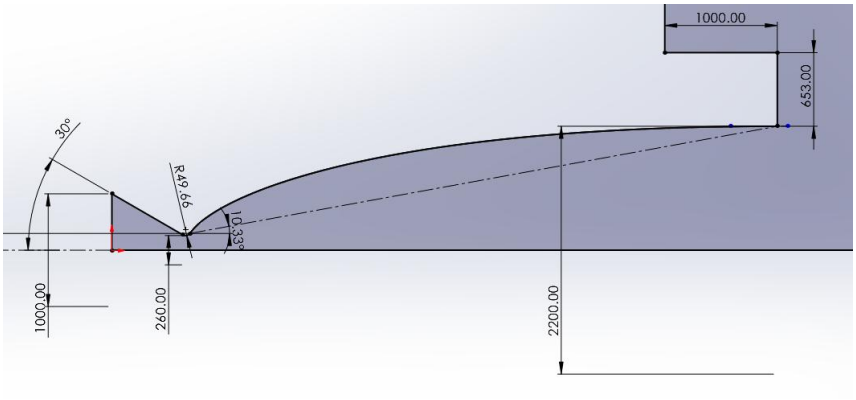


Figure 3:Nozzle Wall Dimensions.

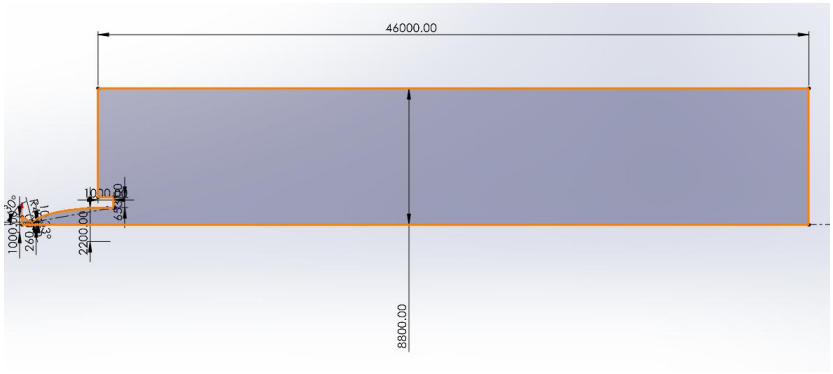


Figure 4:Ambient Dimensions.

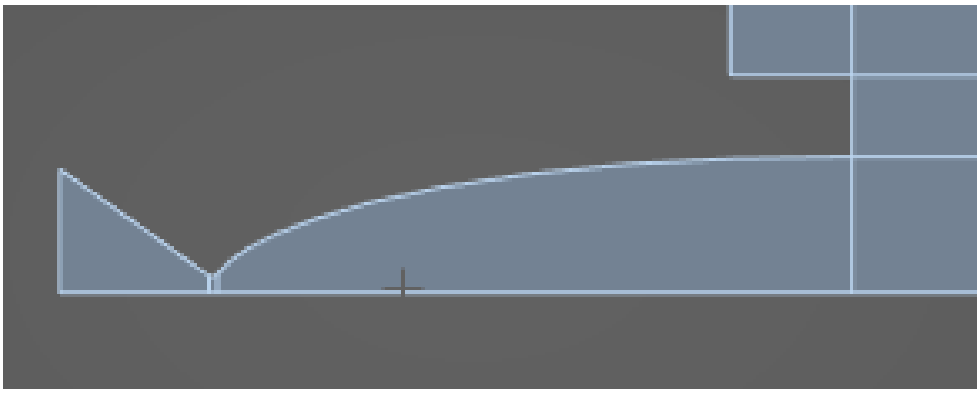


Figure 5:Segmented Nozzle

Turbulence Model and Boundary Conditions

•**Solver**: Axis Symmetric, Pressure Based for initial solution, then Density Based for accurate Mach diamonds Capturing.
•**Turbulence Model**: *K-Omega SST* with *Viscous Heating*
•**Energy**: On
•**Species Transport**: Enabled "H_2O, O_2 and Air"
•**Viscosity**: *Sutherland* for each individual gas and *Mass Weighted Law* for the mixture
•**C_P**: *NASA9 Piecewise Polynomial* for each individual gas, and *Mixing Law* for the mixture.
•**Thermal Conductivity**: *Kinetic Theory* For each gas and *Mass Weighted Law* for the mixture
•**Density**: *Ideal Gas Law.*
•**Method**: *Coupled Scheme* with all parameters set to *Second Order Upwind Scheme* with *Pseudo Time Step* Set to Global Time.
•**URF**: All set to default values, changes were not required during solving.

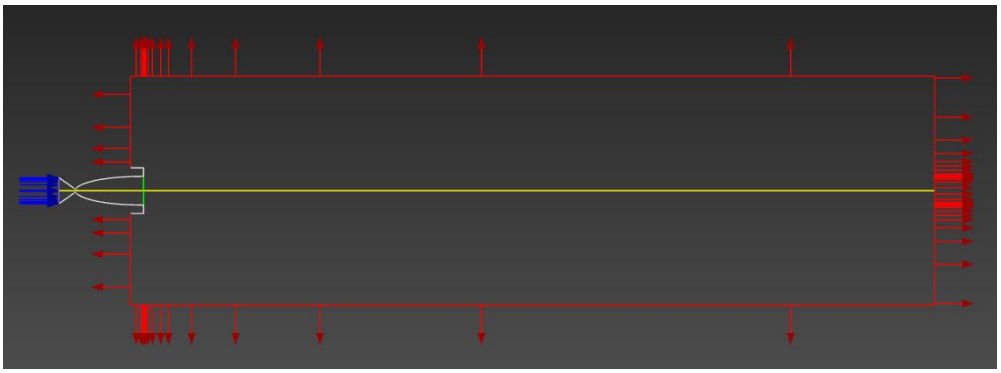


Figure 6:Boundary Conditions

Boundary Conditions:

BC	Pressure (Pa)	Temperature(K)	Mach Number	Wall Shear (Pa)	Species Mass Fraction	Additional
Inlet	12500000	3580	-	-	0.0587H_2(g)+ 0.9413 H_2O(g)	Backflow: Normal To boundary
Outlet	11500	300	0	-	1 Air	Backflow: From Neighboring Cells
Nozzle Wall	-	-	-	No Slip	-	Adiabatic Wall
Step Wall	-	-	-	0	-	Adiabatic Wall
Symmetry	-	-	-	-	-	Axis

Notes

• Outlet was used in radial boundaries to allow flow entertainment caused by the nozzle jet, just like the real-world scenario.
•Ambient Pressure and temperature were set per altitude calculation from NASA website.

Convergence Criterion

•Created Named Selection for the nozzle exit, highlighted in green in figure 5, so I can calculate the key parameters of concern; Axial Velocity, Mass Flowrate, Exit Pressure and Thrust.
•Thrust was calculated via area integral of the momentum of the cell, multiplied by 2π to get the circular cross section integral of the nozzle exit.

$$Thrust = \iint (\rho \cdot u_x^2 + p_x - p_{ambient}) dA_{cell} \cdot 2\pi$$

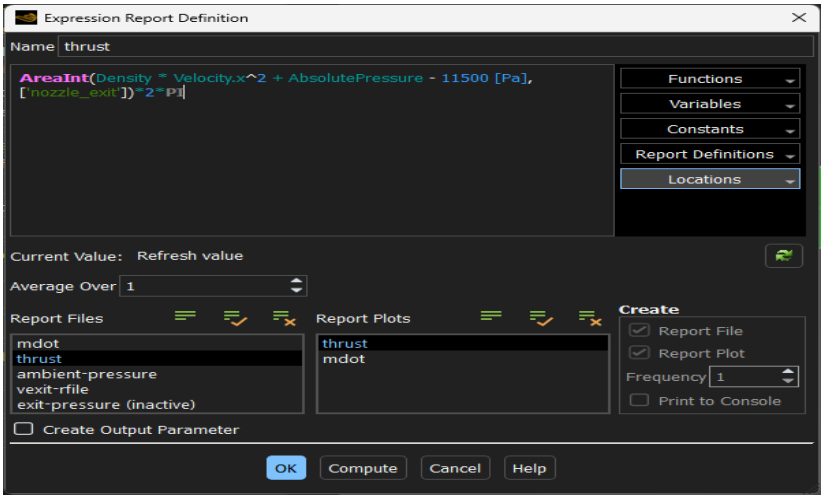


Figure 7:Report Definition for thrust

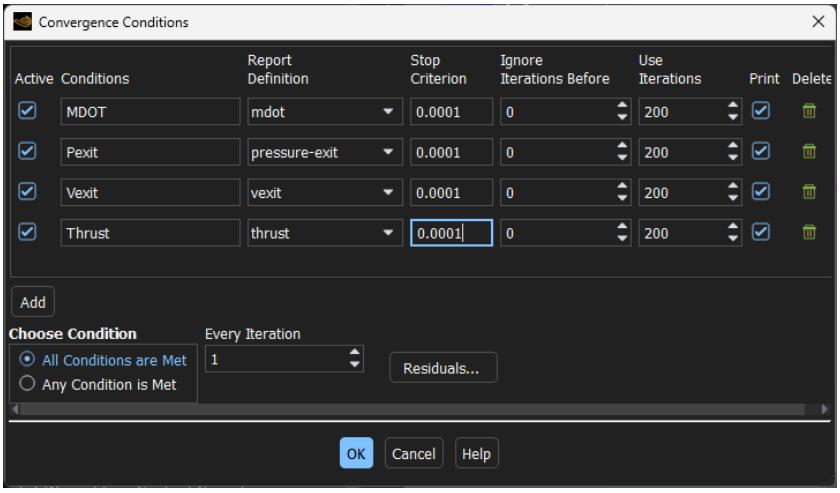
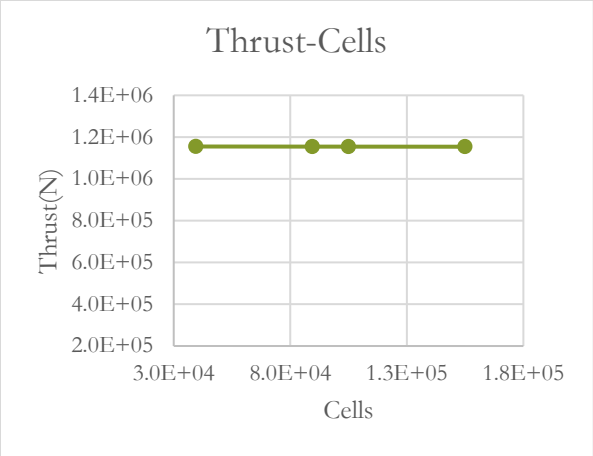
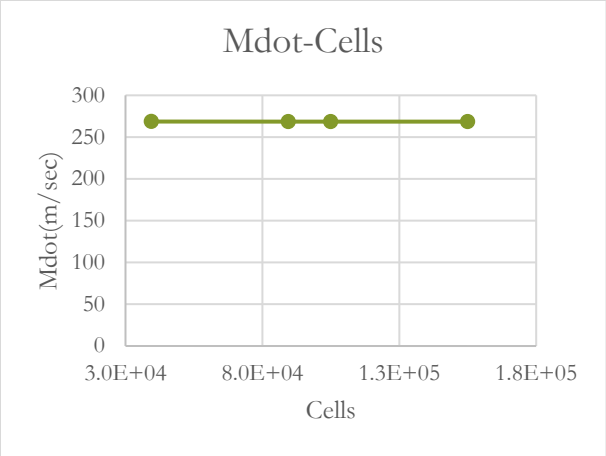
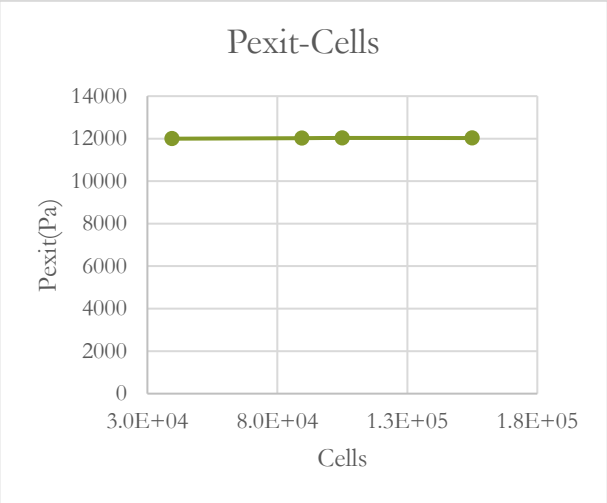
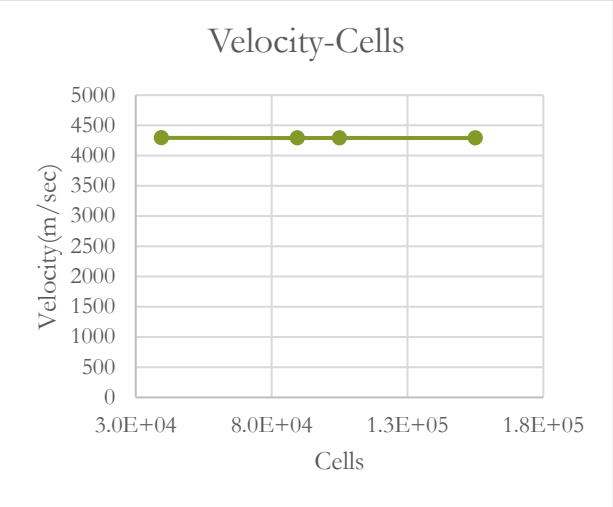


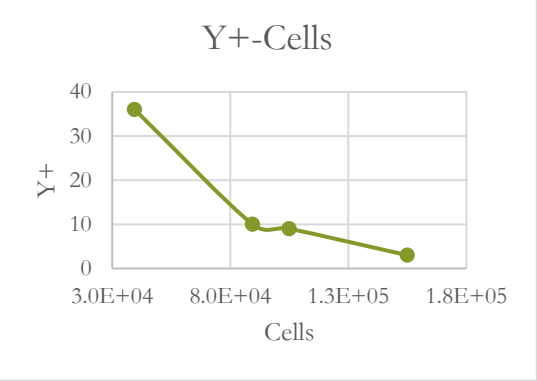
Figure 8: Convergence Criterion











As we can see, the solution did not change much with refinement, with maximum difference of 1.05% between the finest and coarsest meshed, but the maximum Y+ decreased till it reached the recommended value of below 5 at the finest mesh. The maximum Y+ was at the diverging wall, that's because of the analytical calculations using Flat plate theory assumes the wall does not have curvature.

Computational Domain

Mesh generation

•Structured mesh was used due to its efficiency and fast convergence. Special technique was used to create refinement in the approximate location of the Prandtl Meter Expansion Fan without segmenting it.

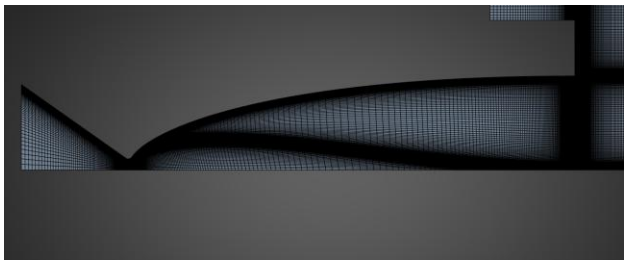


Figure 9:Nozzle Domain

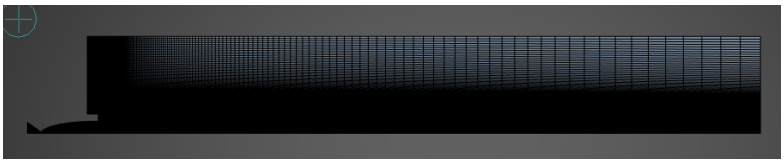


Figure 10:Ambient Domain

Mesh Metric	Skewness
Min	1.3057e-010
Max	0.56866
Average	9.5332e-002
Standard Deviation	0.15894

Mesh Metric	Aspect R...
Min	1.
Max	33130
Average	233.16
Standard Deviation	1333.1

Figure 11: Quality Metrics

•The mesh metrics are giving good sign of a fine mesh, the maximum aspect ratio mesh is located at the fine cells at walls, that is acceptable. The ambient aspect ratio is well below 1000.

Adaptive Mesh Refinement

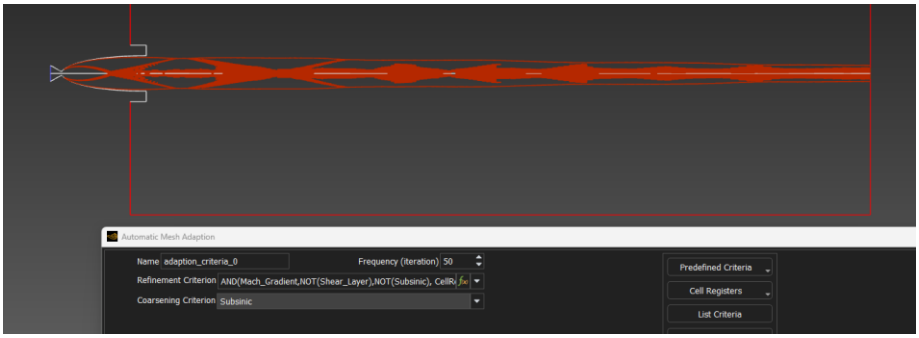


Figure 12:Adaptive Refinement

•Adaptive refinement was used after convergence to sharpen the compression waves. targeting the cells with high Mach number gradient, subtracting the subsonic region and cells with high shear

stress to avoid refining the shear layer. It didn't have any changes on the solution.

Results and post processing

Contours

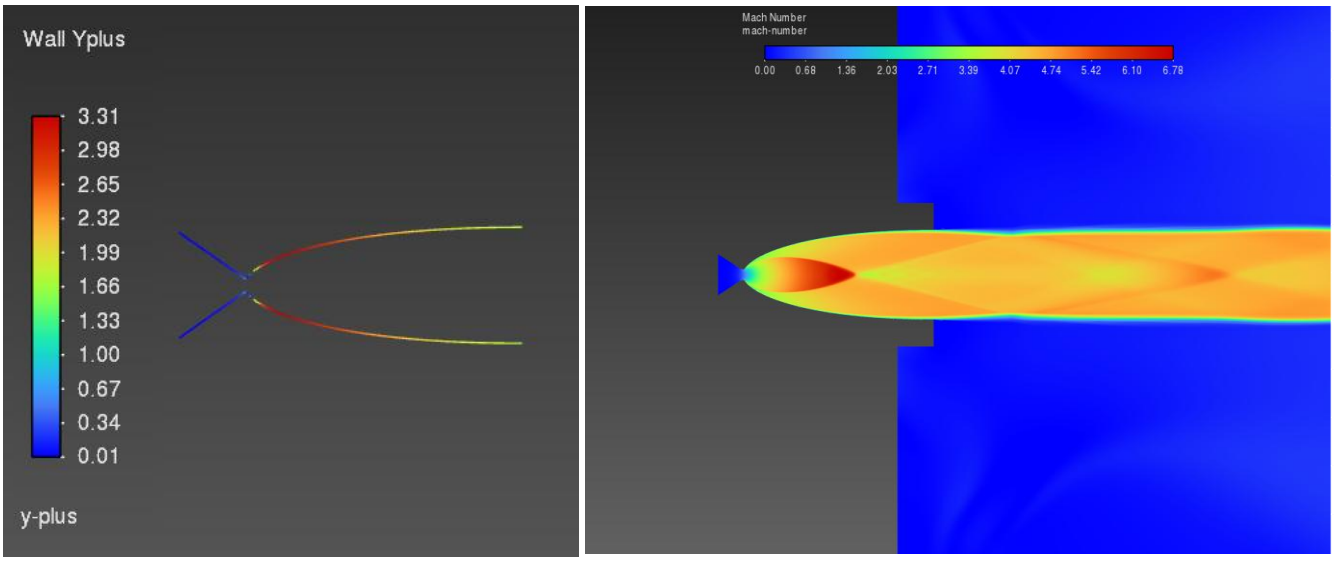


Figure 13: Wall Y+ and Mach Number Contours

•We can see the Prandtl Meyer expansion fan at the throat which is because of the sharp initial expansion angle of 53.3 degrees. Minimum Length Bell Nozzles are famous for this kind of expansion waves to stabilize the pressure at the nozzle exit, there are other types of Bell-shaped nozzles but it is out of the scope of the project.

•We can also see the developing boundary layer captured nicely.

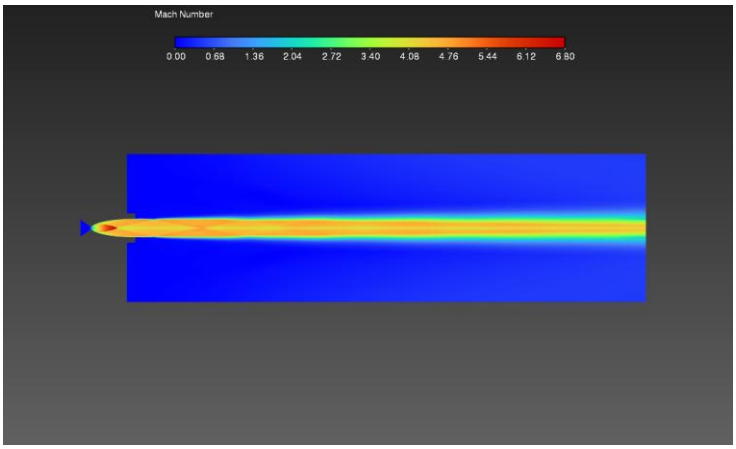


Figure 14: Mach Diamond

The exhaust jet shows a weak compression wave which tapered off close enough after the nozzle, showing how efficient the design is.

• Here's an animation video for the simulation from initialization to full development

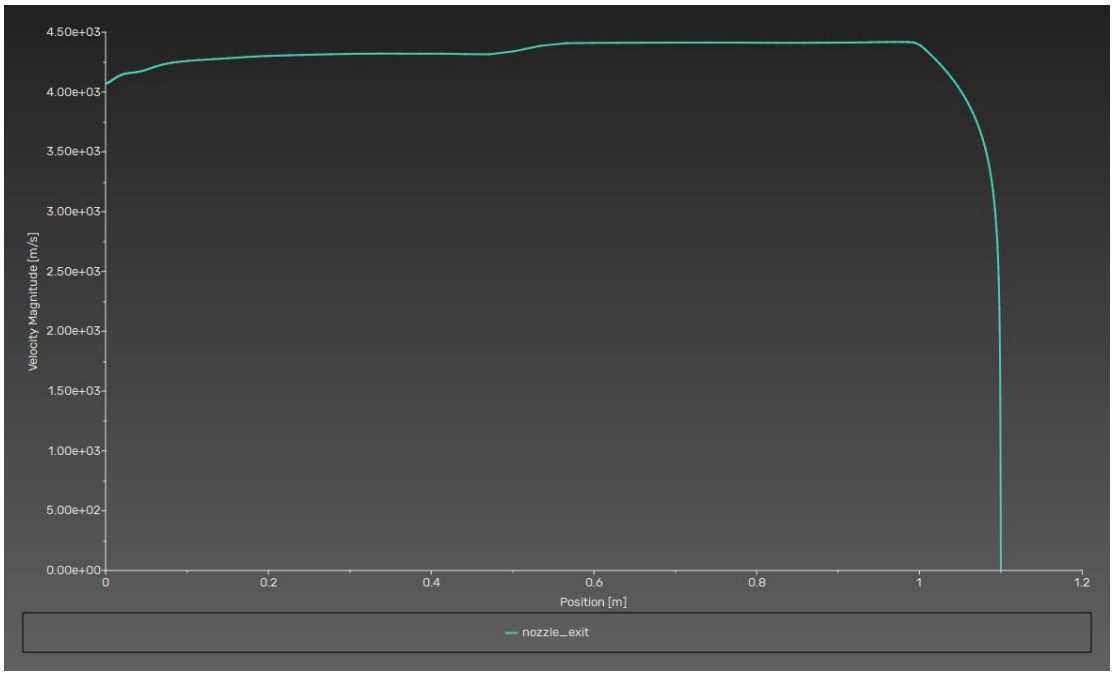


Figure 15:Velocity gradient at exit

The velocity gradient on the exit shows that the boundary layer thickness reduced the effective exit radius to about 1.02m "99% of 4290m/sec freestream", that corresponds to 3.26m^2. that would result to reduction of exit Mach number due to less $\frac{A_{exit}}{A^*}$, increase of exit pressure due to lower expansion ratio, and overall thrust will be reduced, to retrieve the lost thrust, exit radius should be increased by 0.1m.

Comparison with isentropic equation's calculations

Parameter	CFD	Isentropic Equations	Comment
$\dot{m}$(kg/sec)	268.52447	267.8	Less than 1% error.
P_{exit}(Pa)	12027.243	11453	Effective radius of the exit is less than the physical one because of boundary layer thickness. Results in under expanded gases
V_{exit}(m/sec)	4290.3207	4279	Less than 0.26% error.
Thrust(N)	1166532.5	1189205.25	-1.9% apart, typical viscous losses in a nozzle is between 1% and 7%.
Exit temperature (K)	1122.504	1130	-
Exit Mach Number	4.4508288	4.687	Due to Boundary Layer thickness.